KKBOX'S MUSIC RECOMMENDATION SYSTEM

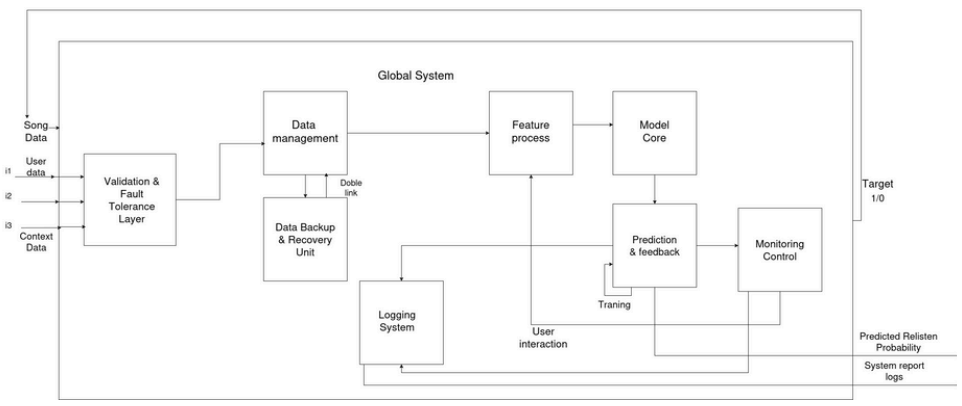
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INTRODUCTION

The present project aims to analyze and design the system related to a music recommendation.

The main objective of the system is to *develop* a *ML model* able to *predict* if a user will repeat a song after hearing it for the first time within a month.

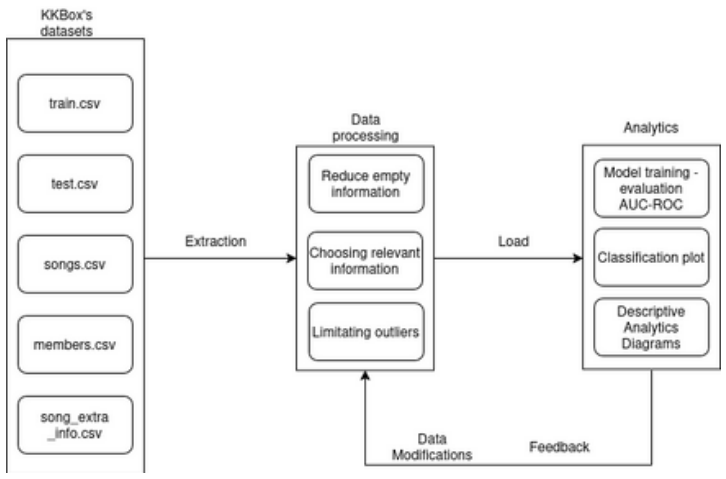


System Architecture Overview

METHODOLOGY

The ETL process extracts multiple datasets and transforms them by cleaning missing/outliers values and selecting relevant features. The processed data is then loaded into the analytics pipeline for model training, evaluation, and descriptive analysis, with feedback used to refine the preprocessing steps

ML Model:
A Random Forest classifier was trained using 8 selected features, consistent categorical encoding, and an 80/20 stratified split. Main evaluation metrics: ROC–AUC and accuracy.



Data Ingestion Flow

Cellular Automata Simulation:
A 50×50 CA with Moore neighborhood was implemented to model local propagation of replay tendencies through probabilistic events (replay activation, neighbor influence, noise, etc.).

RESULTS

Evaluation Results:
The trained model was evaluated using different metrics, such as ROC-AUC curve.
The model showed an acceptable performance with a value of 0.62981, making it better than a random algorithm. (this is the Kaggle punctuaction)

T**he performance by using the full clean dataset shows the next values:**

- ROC–AUC (test full): 0.6634
- Accuracy (full): ~0.63

After reducing the amount of data used for it due to its elevated cost shows the next values ater being tested locally:

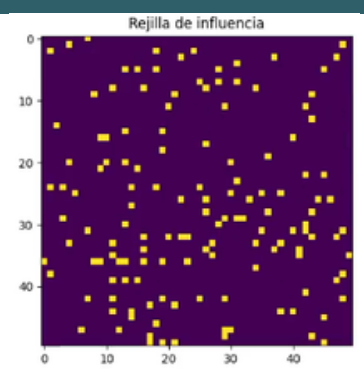
- ROC–AUC (slice 50k): 0.7345
- Accuracy (slice 50k): 0.7435

Finally, the value obtained by the competition after being submitted was:

- Kaggle Scores: 0.62981

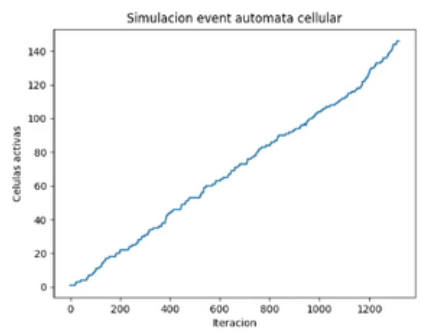
This first iteration establishes a baseline from which more sophisticated feature engineering, model tuning, and alternative algorithms can continue to improve predictive accuracy.

DISCUSSION



The ML model showed strong dependence on data quality, with performance improving significantly after reducing noise and enforcing consistent categorical encoding.

In contrast, the Cellular Automata simulation displayed high sensitivity to small probability changes, revealing unstable but insightful dynamics in replay propagation.



Data about Kaggle model.

CONCLUSION

The ETL and modeling pipeline successfully produced a consistent prediction system, as reflected in performance metrics that align with the Kaggle benchmark. The results highlight the importance of thorough preprocessing for achieving stable learning behavior. While the Random Forest model effectively captured replay patterns, the Cellular Automata simulation revealed the sensitivity of replay dynamics to small local variations.

Overall, the system demonstrates a solid foundation for recommendation tasks, with future improvements centered on advanced models, richer feature engineering, and refined data-quality strategies.